TeamChat specifikacijos įvertinimas ir planuojami įgyvendinimai

1. Specifikacijos vertinimas

Perskaitę specifikaciją susidarėme gerą įspūdį: dokumentas tvarkingas, logiškai išdėstytas ir lengvai skaitomas. Jame yra viskas, ko reikia norint pradėti kurti sistemą: reikalavimai su prioritetais, verslo procesų schemos, nefunkciniai reikalavimai, architektūra, ekranų eskizai ir duomenų modelis. Diagramos ir paaiškinimai yra aiškūs, todėl iš dokumento galima suprasti, kaip sistema turėtų veikti ir iš ko ji susideda.

Kai kur tarp skyrių pastebėjome smulkių neatitikimų, pavyzdžiui, dalis ekranų eskizų ir duomenų modelis ne visai sutampa su reikalavimais. Tai nėra didelės problemos ir manome, kad jas pavyks išspręsti pasitikslinus su autoriais.

2. Reikalavimai, kuriuos manome, kad įgyvendinsime

2.1. Tikrai įgyvendinsime

Tapatybė ir prieiga

FR-01 A user can create an account with email and password, confirmed with a 6-digit code sent by email.
FR-02 A user can sign in with email and password and sign out.
FR-03 A user who forgot their password can reset it through a code sent by email.

Kanalai ir tiesioginės žinutės

FR-09 A user can create a public or private channel with a unique name and an optional description.
FR-10 A user can browse and join public channels.
FR-11 A user can join a private channel only by invitation from a channel member.
FR-12 A user can leave a channel.
FR-13 The channel owner can add and remove channel members.
FR-14 A user can start a DM with any other user of the organisation.

Žinutės

FR-15 A member can send a text message to a channel or DM, it appears to all online members in real time.
FR-16 A user sees the message history of a conversation, newest first, and can load older messages.

Failai

FR-23 A member can upload files to a conversation (up to 10 files per message, 25 MB each). Each file is stored once in file storage, and the message contains a link to it.
FR-24 Members of the conversation can download a shared file, non-members cannot.

Prisijungimo būsena

FR-32 Every user’s status (Online, Away, Offline) is shown next to their name and updated automatically.

Darbo eigos (workflow)

FR-34 A channel member can create a workflow in the channel with a title and description.
FR-35 A workflow can be assigned to a channel member, or a member can claim it with “On It”.
FR-36 The assignee can move the workflow through the stages To Do – In Progress – Done.

Pranešimai

FR-39 A user receives notifications for new DMs, mentions, channel invitations and workflow changes. The latest are shown on the main page.

2.2. Įgyvendinsime, jei pakaks laiko

FR-04 A user can edit their profile: display name and profile picture.
FR-05 A user can change their password while signed in.
FR-17 A user sees which conversations have unread messages.
FR-18 A user can edit and delete their own messages, an administrator can delete any message.
FR-19 A user can mention another member with ‘@name’, which notifies them.
FR-25 A user can see all files shared in their conversations on the Files page.
FR-33 A user can set their status manually (e.g. Away).
FR-37 A user sees all workflows assigned to them on the Workflows page.
FR-40 A user can search messages and file names in the conversations they belong to.

2.3. Papildomi

FR-06 A user can export their data and delete their account.
FR-38 A workflow can have a due date.

FR-26 Images are shown as a preview in the conversation.
FR-07 An administrator can deactivate a user account.

2.4. Neplanuojame įgyvendinti

FR-08 Sign-in with external providers (Google, Microsoft).
FR-20 A user can react to a message with an emoji.
FR-21 A user can reply to a message in a thread.
FR-22 Typing indicator ("Anna is typing…").
FR-41 Email notifications for unread messages while offline.
FR-27 A user can start an audio call from a DM, the other user receives an incoming call and can accept or decline.
FR-28 A participant can turn the video and microphone on and off and leave the call.
FR-29 A member can start a call in a channel and up to 4 members can join it.
FR-30 Screen sharing during a call.
FR-31 Call recording.

3. Nefunkciniai reikalavimai

SEC-01 Password storage: Argon2id or bcrypt hashes; passwords are never stored or logged in plain text; minimum 8 characters with at least one letter and one digit.
SEC-02 Transport security: HTTPS and WSS (TLS 1.2 or higher) everywhere.
SEC-03 Tokens: access token lifetime ≤ 15 min; refresh token 7 days, rotated on every use and revoked on sign-out or password change. The WebSocket is authenticated during the handshake with a short-lived, single-use ticket.
SEC-04 One-time codes: 6-digit codes for email confirmation and password reset; valid 10 min; at most 5 wrong attempts; resend at most once per 60 s; stored hashed.
SEC-05 Authorisation: every REST request and every WebSocket subscription checks membership. Non-members never receive content of a private channel or DM.
SEC-06 Input handling: server-side validation of all input, parameterised queries, escaping of message text against XSS, CSRF protection if cookies are used.
SEC-08 File upload safety: max 25 MB; allow-list of file types checked on content, not only on extension; random storage names; files never served from the web root and always downloaded with Content-Disposition: attachment.
REL-02 No message loss: once the server confirms a message to the sender, it is persisted and survives a server restart.
REL-03 Idempotent sending: the client generates a clientMessageId; repeating a request with the same id returns the original message and never creates a duplicate.
REL-04 Ordering: all participants see the messages of a conversation in the same order, defined by the server-assigned per-conversation sequence number seq.
REL-05 Reconnection: after a lost connection the client reconnects automatically with exponential backoff (1 s up to 30 s), then fetches missed messages and notifications through REST using its last-seen cursor.
PERF-03 History loading: the first page of 50 messages is shown within 1 s; older messages load with cursor pagination, 50 per page.
PERF-07 File transfer: uploads and downloads are streamed in 64 KB chunks, so server memory per transfer stays at one 64 KB buffer whatever the file size.
COMP-02 Server delivery: runs on Linux as containers (application server, PostgreSQL), started with one command (docker compose up); configuration only through environment variables.
MAIN-03 Schema migrations: versioned database migrations (e.g. Flyway or Liquibase); no manual schema changes.